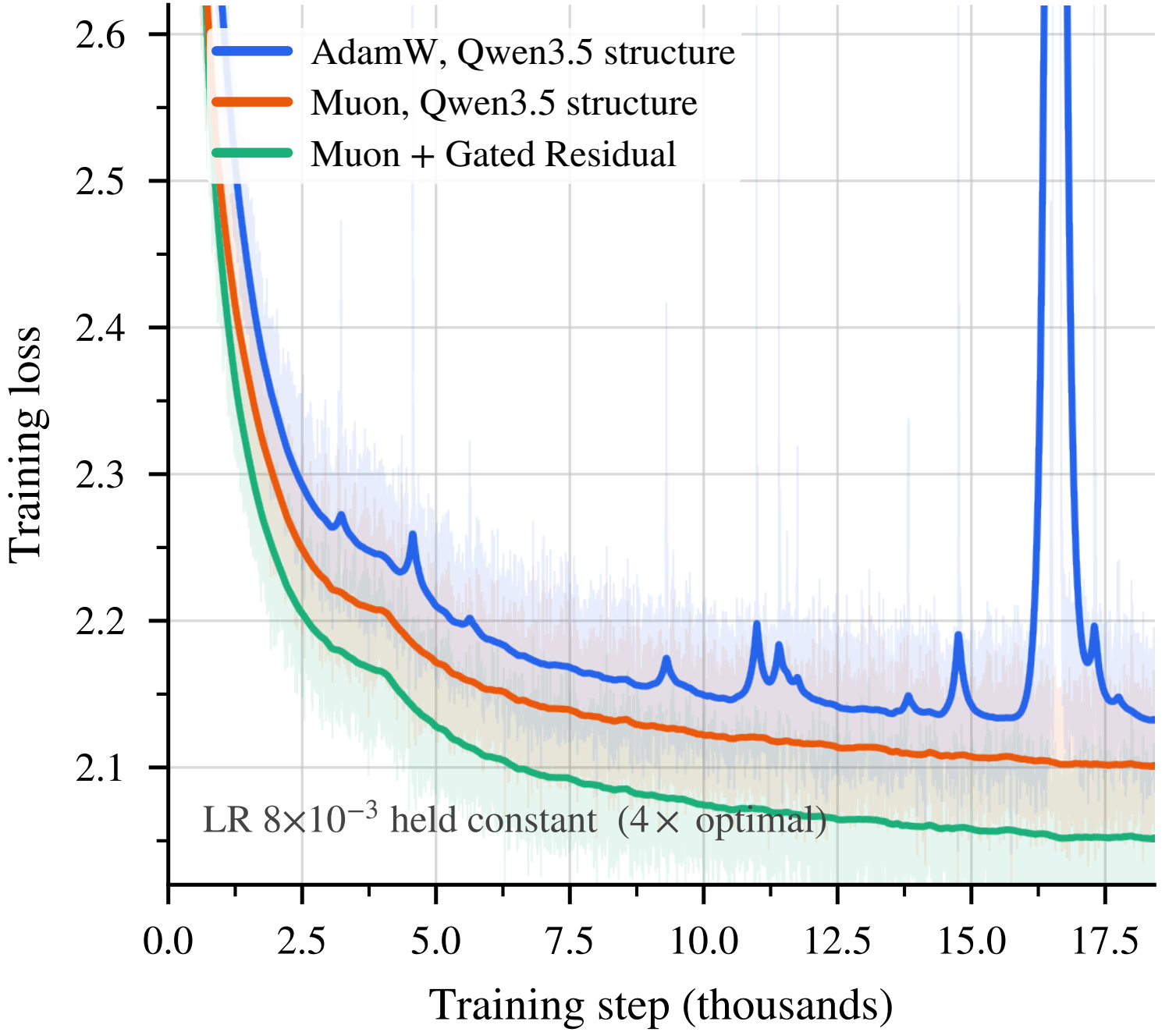

AdamW, Qwen3.5 structure
Muon, Qwen3.5 structure
Muon + Gated Residual
LR 8×10⁻³ held constant (4× optimal)
Training loss
2.6
2.5
2.4
2.3
2.2
2.1
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Training step (thousands)